LESSON 5.2

Determine the Area of Triangles

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the area of a triangle using the formula A = ½ × base × height.

WHY THIS MATTERS Your architecture firm is designing a park with triangular garden beds. The client wants to know exactly how much soil to order, which means calculating the area of each triangular section. The first garden has a base of 12 feet and a height of 8 feet.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Base *Spanish: Base*	Any side of a triangle can be the base. Once you pick a base, the height must be measured perpendicular (at a 90° angle) to it.	b in A = ½bh
Height *Spanish: Altura*	The straight-up distance from the base to the top corner.	h in A = ½bh
Area *Spanish: Área*	How much space is inside a flat shape.	measured in cm², in², ft²
Perpendicular *Spanish: Perpendicular*	Two lines that meet to make a square corner (90 degrees).	The corner of a book or a door frame — the edges meet at exactly 90°, shown by a small square symbol
Composite figure *Spanish: Figura compuesta*	A shape made by putting two or more simple shapes together.	A house shape = a rectangle (the walls) + a triangle (the roof); total area = rectangle area + triangle area
Formula *Spanish: Fórmula*	A math rule written with symbols.	A = ½ × b × h means Area equals one-half times base times height; for b = 12 and h = 8, A = 48

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A triangle has a base of 14 cm and a height of 9 cm. Find its area.

1. 14 × 9 = 126
2. 126 is the parallelogram made by two triangles.
3. 126 ÷ 2 = 63

Answer: 63 cm²

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now a smaller triangle: base = 10 cm, height = 6 cm.

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The base of the triangle is _____ and the height is _____ .
- I know that measure is the height because _____ .
- Two copies of this triangle would compose a _____ .
- Area = ½ × _____ × _____ = _____ square units.

Word bank triangle | parallelogram | half | base | height | perpendicular | square corner | area | formula | right triangle | copy | rotate

WATCH OUT!

A common mistake

A common mistake in Area of Triangles is forgetting the ½

LESSON 5.2

Determine the Area of Triangles

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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Answer: 63 cm²

1 **Circle the letter of the best answer. Show how you know.**

What is the area of a triangle with base 10 cm and height 6 cm?

(A) 16 sq cm
(B) 30 sq cm
(C) 30 cm
(D) 60 sq cm

STEPS TO TRY

1. You have a base of 10 cm and a height of 6 cm — a triangle needs only those two, plus one special step.
2. A triangle is half of a rectangle: use A = ½ × b × h. If you only multiply, your answer is twice too big.
3. Multiply 10 × 6, then take half of the product. Area units are square centimeters.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

2 **Circle the letter of the best answer. Show how you know.**

A triangular garden bed in the firm's park design has an area of 24 sq ft and a base of 8 ft. What is the height?

(A) 3 ft
(B) 6 ft
(C) 12 ft
(D) 16 ft

STEPS TO TRY

1. The area (24 sq ft) and base (8 ft) are given — you're solving backwards for the height.
2. Simplify first: ½ × 8 = 4, so the equation becomes 24 = 4 × h.
3. Divide 24 by 4 to find the height in feet.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

3 **Circle the letter of the best answer. Show how you know.**

Why do we divide by 2 when finding the area of a triangle?

(A) A triangle is exactly half of a rectangle with the same base and height
(B) Triangles have 2 equal sides
(C) The base is always twice the height
(D) We always divide area formulas by 2

STEPS TO TRY

1. Think about drawing a diagonal across a rectangle — what two shapes does it create?
2. Compare a triangle to a rectangle with the same base and height. How much of the rectangle does the triangle cover?
3. Sketch a rectangle, draw one diagonal, and shade one triangle. What fraction of the rectangle did you shade?

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

What is the area of a triangle with base 14 ft and height 6 ft?

(A) 20 sq ft
(B) 42 sq ft
(C) 42 ft
(D) 84 sq ft

STEPS TO TRY

1. Base 14 ft and height 6 ft — remember what makes a triangle's area different from a rectangle's.
2. Use A = ½ × b × h: compute the full product first, then halve it. Area needs square units.
3. Find 14 × 6, then take half of that product. Write the units as sq ft.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

EXPLAIN YOUR THINKING

The triangle fits inside a 12 × 8 rectangle. How does the triangle's area compare to the rectangle's area?

SENTENCE STARTER The area of the triangle is ___ the rectangle because A = 1/2 × ___ × ___ = ___ square feet.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help